

Assessment Purpose

Congratulations on passing phase one of the MSA program!

To pass Phase 2, you must complete either the *software* stream or *data science* stream. This assessment is for the *software* stream. This is an **individual assessment**. You will be ranked against your peers.

Building on our foundational knowledge of React and REST APIs using the .NET framework from Phase 1, this assessment will challenge you to develop a full stack application with advanced features. This project aims to demonstrate your proficiency in these technologies, as well as your ability to independently research and integrate new technologies.

AI Usage

Artificial Intelligence (AI) tools such as GitHub Copilot, ChatGPT, Microsoft Copilot, Claude, and similar technologies may be used throughout this assessment.

AI is becoming an essential part of modern software development, and students are encouraged to use it to support research, planning, coding, testing, debugging, and documentation. However, students are expected to critically evaluate AI-generated outputs and demonstrate an understanding of the solutions they implement.

As part of your submission, you must briefly describe how AI was used during the development process. Effective and responsible use of AI will form part of the overall assessment. (*For more details refer to the submission section*)

Task

In this assessment you are to develop a full stack web application. The application **must** be based on the theme mentioned in the following section.

All the basic requirements are required. Additionally you are to achieve at minimum three advanced requirements.

Theme - Gamification

The theme for this year's assignment is **Gamification**. In **Human-Computer Interaction (HCI)**, gamification is the practice of integrating game design elements—such as points, badges, achievements, streaks, leaderboards, and progress tracking—into non-game applications to improve user engagement, motivation, and overall user experience. This does not mean you have to create a game.

One example could be a study app that rewards users for completing tasks and maintaining learning streaks. Students are encouraged to apply HCI and gamification principles creatively in their software solution. Originality and innovation will be valued.



Basic Requirements

A frontend and backend are both required. Your application should contain all these basic requirements and follow the theme; neglecting any of these basic requirements will **result in an instant failure**. Additionally you **MUST** include a README mentioned in the submission section.

Frontend:

- Built using React with TypeScript.
 - (You may use JavaScript, but TypeScript is preferred)

Visually appealing and responsive UI. (Web app displays nicely on both computer and mobile).

- If your web app is designed only for desktop, justify why your web app is not responsive in your readme.
- You may use a styling library such as MUI, Mantine, Tailwind, etc., or use custom styling as long as you focus on a visually appealing design. Your UI should reflect your own design choices and have a unique visual identity.
- Navigation using React Router or a similar routing library.
- Clear use of Git with a **regular commit** history
- Deploy your frontend
- Implement unit tests covering key frontend components and functionality.

Backend:

- Built using C# with **.NET 10 or higher**.
- Must use Entity Framework Core (EF Core).
- Data persistence using an SQL or NoSQL database.
- At minimum, implement CRUD operations (Create, Read, Update, Delete)
- Clear use of Git with a regular commit history
- Implement unit tests covering key backend components and functionality.
- Deploy your backend.
- Expose Scalar API documentation UI instead of Swagger UI.

Advanced Requirements

Your application should contain at least **three** advanced requirements from this list.

Ensure that the advanced requirements you choose are listed on the repo's README or they will **not be marked**. We will only mark three features.

- Integrate all UI components with Storybook.
- Implement Security Measures: Minimum two with justification on their importance in your application.
 - Authorisation (RBAC), Anti-CSRF measures, Password hashing, Data validation / sanitisation, Rate limiting.
 - In the readme include a writeup about their importance and how it was implemented.
- Use a state management library, e.g., Zustand, Redux.
- Support for theme switching (e.g., light/dark mode).
- Dockerize your project using docker.
- Implement WebSockets.
- End-to-end testing using Cypress.
- Performance tests, System Logging and Metrics.
- Multiplayer functionality support
- Caching strategy and API optimization techniques



Submission Instructions:

Please submit using the MSA 2026 Phase 2 submission form located in the Phase 2 GitHub repository. This will be a Microsoft Form. Expect to provide the following

- Link to your public Github repository containing the frontend and backend
- Link to your video. Ensure the link is public!
- Textfield for any secrets or information required for marking you do not want public (Optional)

Repository Content:

- Frontend and backend code
- **README** file containing the following information:
 - Link to your deployment
 - Brief introduction to your project
 - Section explaining how your project relates to the theme.
 - Section explaining what interesting features makes your project unique from others, and worth highlighting to the marker.
 - A clear checklist of advanced features you've implemented.
 - **Important:** We will only mark the top 3 advanced features that are explicitly listed in your README.
 - Self-reflection: If you were to do the project again, what would you do differently
- “/specs” folder containing .md files
 - Contains **evidence of planning, design, and AI-assisted development**
 - Includes AI prompt files (.md), agent instructions, and context/config files
 - Must include prompts used during development (not just final code) written down in (.md) format
 - If you do not have .md files, ask your AI agent to create these files

Video Content:

- Max video length: **6 minutes** (Longer videos will be penalized)
- Part 1: Explain and show how you used AI during your development
- Part 2: Explain some design decisions you made during your project

Unsure how to get started?

Check out the Getting Started exemplar in the software development folder of the MSA Phase 2 GitHub repository. Additionally AI is your best friend in Software development. It is an important tool to use in the industry.

Marking Criteria

A formal marking criteria is not released to the public. The MSA marking team will have a set marking criteria we will follow. Each criteria we will give a score of 0 to 5.

As a general idea, the marking criteria will contain the following:

- Does your application look visually appealing?
- How is your code quality? Is it well structured and easily understood by other developers?
- Git usage. Is there evidence you have commits and have been working on your project? It does not look great if we see one commit with the entire assessment at the end.
- Presentation quality. Were the points discussed in a clear manner? Were you speaking clearly? Is it in a clear structure?
- AI Usage. How effectively were you using AI based on the evidence you have provided in the video and “/specs” folder

Frequently Asked Questions:

Q: Can I use any platform to deploy my frontend and backend?

Yes. However make sure the deployment is accessible until the results of phase 2 have been released.

Q: For the backend can I use another language/framework like Java and Spring Boot?

No. You are required to use C# and the .NET framework

Q: I can't manage to implement the minimum advanced requirements! Am I guaranteed to fail!?

No. We have a marking criteria that every student will go through. As long as you hit the right criteria, it's still possible to pass.

Q: Do I own the project I made?

No.

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Just kidding, of course! It'll be a great project to put on your CV or portfolio.

Q: Can i submit two github links, one for frontend and backend?

No. You can only submit one github link containing to the your repository. This one repository should contain both your frontend, backend and any other necessary files.

Q: Will my commit history be checked?

Yes and may be part of the marking criteria.

Q: What happens if i make a commit after the deadline?

Do not commit after the deadline. Last commits are checked and may lead your submission being disqualified or penalized.



Good luck!

